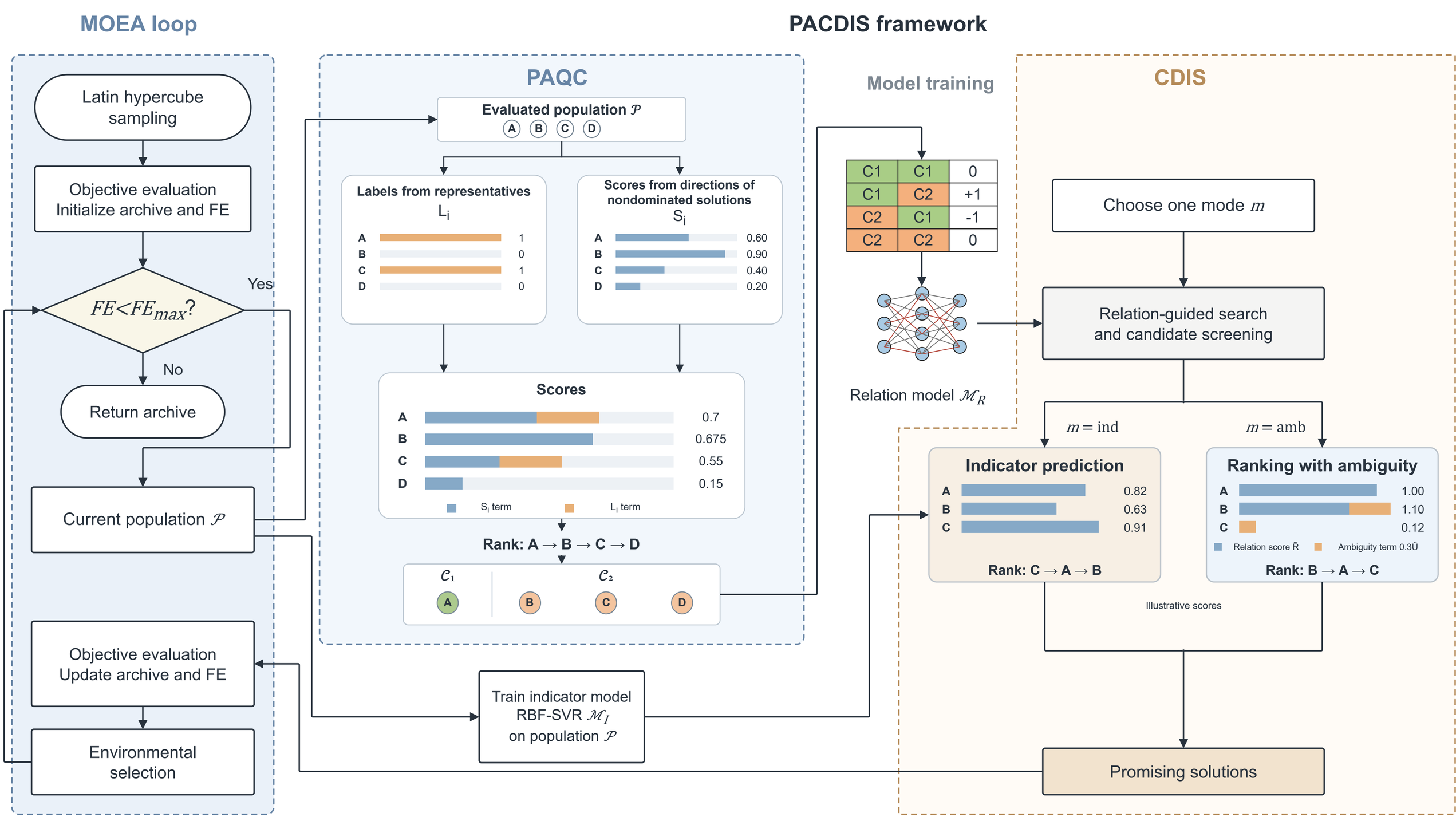

MOEA loop
PACDIS framework
Latin hypercube sampling
Objective evaluation
Initialize archive and FE
FE<FEmax?
Yes
No
Return archive
Current population 𝒫
Objective evaluation
Update archive and FE
Environmental selection
PAQC
Evaluated population 𝒫
A B C D
Labels from representatives
Li
A 1
B 0
C 1
D 0
Scores from directions of nondominated solutions
Si
A 0.60
B 0.90
C 0.40
D 0.20
Scores
A 0.7
B 0.675
C 0.55
D 0.15
Si term
Li term
Rank: A → B → C → D
𝒞1
𝒞2
A
B
C
D
Train indicator model
RBF-SVR ℳI
on population 𝒫
Model training
C1 C1 0
C1 C2 +1
C2 C1 -1
C2 C2 0
Relation model ℳR
CDIS
Choose one mode m
Relation-guided search
and candidate screening
m = ind
m = amb
Indicator prediction
A 0.82
B 0.63
C 0.91
Rank: C → A → B
Ranking with ambiguity
A 1.00
B 1.10
C 0.12
Relation score R̃
Ambiguity term 0.3Ũ
Rank: B → A → C
Illustrative scores
Promising solutions